

Robust Image Classification under Distribution Shift using Deep Learning

Sunny Kumar, Abhishek Kumar Priyadershi, Sumit Chouhan, Priyanshu Kumar Chaudhary, Sahil Sangwan
Indian Institute of Technology Kanpur
Email: [optional]

Abstract—This project addresses the ShiftGuard10 robust image classification task, where the objective is to classify 32×32 RGB images into 10 classes under distribution shift conditions. In this setting, the test data distribution differs from the training data, making generalization challenging.

A custom convolutional neural network incorporating residual connections and Squeeze-and-Excitation (SE) blocks was developed and trained from scratch. Advanced preprocessing and training strategies including MixUp, CutMix, Stochastic Weight Averaging (SWA), and Stratified K-Fold Cross Validation were employed to improve robustness.

The final model achieved a cross-validation F1 score of approximately 0.86 and a Kaggle leaderboard score of around 0.79, demonstrating effective generalization under distribution shift.

Index Terms—Image Classification, Deep Learning, Distribution Shift, CNN, Ensemble Learning

I. INTRODUCTION

Image classification models often perform well on standard datasets but struggle when exposed to unseen data distributions. This issue, known as distribution shift, is common in real-world applications.

This project focuses on developing a robust model that generalizes effectively under such conditions using advanced deep learning techniques.

II. OBJECTIVE

The primary objective of this project is to develop a robust deep learning model capable of accurately classifying 32×32 RGB images into 10 classes under distribution shift.

The key goals are:

- Maximize macro F1-score
- Improve generalization on unseen data
- Train the model from scratch
- Ensure reproducibility

III. PROBLEM DESCRIPTION

The dataset consists of 32×32 RGB images across 10 classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck.

A. Evaluation Metric Justification

In this work, the macro F1-score is used as the primary evaluation metric instead of accuracy. Accuracy measures the overall proportion of correct predictions but can be misleading in scenarios involving class imbalance or varying difficulty across classes.

The dataset used in this project contains classes with varying levels of difficulty and potential imbalance. In such cases, a model may achieve high accuracy by performing well on easier or more frequent classes while performing poorly on harder or less frequent ones.

The macro F1-score addresses this limitation by computing the F1-score independently for each class and then averaging them. This ensures that all classes are treated equally, regardless of their frequency.

Additionally, F1-score balances both precision and recall, making it more suitable for evaluating model performance under distribution shift, where both false positives and false negatives are important.

Therefore, macro F1-score provides a more reliable and fair evaluation of model performance compared to accuracy in this setting.

Constraints:

- No pretrained models: The model must be trained entirely from scratch without leveraging external learned features.
- No external datasets: Only the provided dataset can be used, ensuring a fair and controlled evaluation.
- Fully reproducible notebook: The complete pipeline must be reproducible, allowing consistent results across different runs.

Evaluation Metric: Macro F1-score

IV. PREPROCESSING

- Normalization: Standardized pixel values to ensure stable gradients and faster convergence.
- Random cropping with padding: Improved spatial invariance by exposing the model to shifted inputs.
- Horizontal flipping: Enhanced robustness to orientation variations in input images.
- TrivialAugmentWide: Introduced diverse transformations to increase data variability and prevent overfitting.

Additionally:

- MixUp: Combines images and labels
- CutMix: Replaces image regions

These techniques help reduce overfitting and improve generalization.

V. MODEL ARCHITECTURE

The proposed model is based on a Convolutional Neural Network (CNN) architecture enhanced with residual connec-

tions and Squeeze-and-Excitation (SE) blocks. This combination enables effective feature extraction, stable training of deep networks, and improved attention to important features.

A. Convolutional Neural Network (CNN)

The core of the model consists of convolutional layers that act as feature extractors. These layers apply learnable filters to the input images to capture spatial patterns. In the initial stages, the network learns low-level features such as edges, textures, and color gradients. As the depth increases, the model progressively learns higher-level semantic features such as shapes and object structures.

Pooling operations are used to reduce spatial dimensions while retaining important information, which helps in reducing computational complexity. This hierarchical feature extraction allows the model to build a rich representation of the input data.

B. Residual Learning

To enable deeper network design without degradation in performance, residual connections are incorporated. Instead of directly learning a mapping from input to output, the network learns a residual function, where the input is added to the output of intermediate layers through shortcut connections.

This approach helps in addressing the vanishing gradient problem, allowing gradients to propagate more effectively during backpropagation. As a result, the network can be trained more efficiently and achieves better convergence. Residual connections also help in preserving important low-level information across layers.

C. Squeeze-and-Excitation (SE) Mechanism

To further enhance feature representation, Squeeze-and-Excitation (SE) blocks are integrated into the architecture. These blocks introduce a channel-wise attention mechanism that allows the network to adaptively recalibrate feature maps.

First, global average pooling is applied to compress spatial information into channel descriptors. These descriptors are passed through fully connected layers and non-linear activations to learn channel dependencies. A sigmoid function generates weights for each channel, which are then used to scale the feature maps.

This process enables the model to emphasize important feature channels while suppressing less informative ones, thereby improving discriminative power and overall performance.

VI. TRAINING METHODOLOGY

A. Data Augmentation

- Random cropping with padding: Improves spatial invariance by exposing the model to shifted inputs.
- Horizontal flipping: Enhances robustness to left-right orientation variations.
- TrivialAugmentWide: Introduces diverse transformations to prevent overfitting.
- Normalization: Ensures stable gradients and faster convergence.

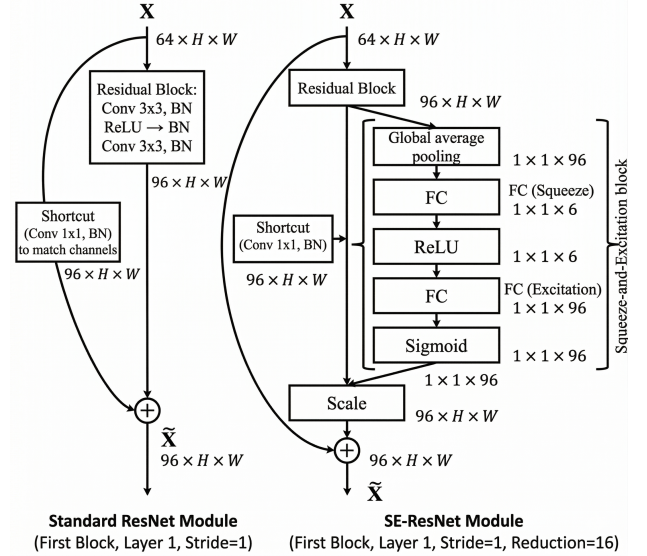


Fig. 1. Residual and SE block architecture.

Advanced techniques:

- MixUp: Encourages linear behavior between samples by interpolating inputs and labels.
- CutMix: Promotes region-level feature learning by mixing image patches.

B. Training Strategy

The model was trained using a 5-fold stratified cross-validation approach to ensure balanced class distribution across all folds and to obtain reliable performance estimates. This strategy helps reduce variance and improves the robustness of the evaluation. During training, advanced data augmentation techniques such as MixUp and CutMix were applied to enhance generalization and mitigate overfitting. Automatic Mixed Precision (AMP) was utilized to accelerate training and reduce memory usage, while maintaining numerical stability.

C. Optimization

The model was trained using stochastic gradient descent (SGD) with Nesterov momentum to achieve stable and efficient convergence. A OneCycle learning rate scheduler was employed to vary the learning rate dynamically, enabling faster convergence and improved generalization. Weight decay was incorporated as a regularization mechanism to reduce overfitting, while gradient clipping was applied to control gradient magnitude and ensure stable training.

D. Advanced Techniques

- Automatic Mixed Precision (AMP)
- Stochastic Weight Averaging (SWA)

VII. RESULTS

A. Cross-Validation Performance

- Fold 1: 0.8512
- Fold 2: 0.8530

- Fold 3: 0.8760
- Fold 4: 0.8684
- Fold 5: 0.8685

Average F1 Score $\approx$ 0.863

B. Kaggle Performance

Public Leaderboard Score $\approx$ 0.79

VIII. ABLATION STUDY

To evaluate the contribution of different components, key techniques were removed individually and their impact was observed.

Removing MixUp and CutMix led to increased overfitting and reduced generalization. Without SWA, the model showed less stable validation performance across epochs. Excluding ensemble learning resulted in a performance drop of approximately 2–3%.

These results indicate that augmentation, SWA, and ensembling each play an important role in improving robustness under distribution shift.

IX. DISCUSSION

The results demonstrate that standard training approaches are insufficient for handling distribution shift. Techniques such as MixUp, CutMix, and SWA significantly improved generalization by reducing overfitting and stabilizing training.

Ensemble learning further enhanced performance by reducing prediction variance across folds. However, a gap between cross-validation and leaderboard performance indicates the presence of a distribution mismatch between training and test data.

Overall, the combination of augmentation, optimization strategies, and ensembling was crucial in achieving robust performance.

X. CONCLUSION

In this work, a robust image classification model was developed to address distribution shift in the ShiftGuard10 dataset. By combining advanced data augmentation techniques, a custom CNN architecture, and training strategies such as SWA and ensemble learning, strong performance was achieved.

The model obtained a cross-validation F1 score of approximately 0.86 and a Kaggle score of around 0.79, demonstrating effective generalization. The results highlight the importance of integrating multiple techniques to handle real-world challenges in machine learning.

XI. FUTURE WORK

Future work includes exploring deeper architectures such as ResNet34 or EfficientNet, improving data augmentation strategies, and performing systematic hyperparameter tuning. Additionally, semi-supervised learning techniques can be investigated to further improve performance under distribution shift.

ACKNOWLEDGMENT

The authors would like to thank the course instructor and teaching assistants for their guidance and support throughout this project. We also acknowledge the Kaggle platform for providing the dataset and evaluation framework.

REFERENCES

- [1] K. He et al., “Deep Residual Learning for Image Recognition”
- [2] PyTorch Documentation
- [3] Kaggle ShiftGuard10 Competition